Autoreason: Autoresearch for Subjective Domains

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https://github.com/NousResearch/autoreason

Abstract. We introduce autoreason, a method for iterative LLM refinement in subjective domains where no objective metric exists. The method constructs a subjective fitness function through independent blind evaluation, extending the autoresearch paradigm. Autoreason uses fresh isolated agents per role, blind judge panels with Borda count aggregation, and conservative tiebreaking. Across 5 task types with claude-sonnet-4, autoreason ranked first or second in every 7-judge blind comparison, winning 3 of 5 tasks outright (avg Borda 27.8/35, avg rank 1.4). Chain-of-thought judging reduces convergence time by 3×. Monte Carlo analysis shows 80% convergence rate with consistent output quality. Retroactive game-theoretic analysis confirms fully transitive preferences and Elo plateau, validating the convergence mechanism. We find the method's advantage is task-dependent and iteration-dependent: autoreason excels on tasks with genuine tradeoffs, while a single autoreason pass does not outperform simpler baselines. A scaling experiment with a stronger model (Sonnet 4.6) reveals a failure mode where synthesis consistently outcompetes the incumbent, preventing convergence entirely. A margin requirement—challengers must win by ≥2 Borda points to displace the incumbent—recovers convergence (15–17 passes), while scope-aware judging and plateau detection do not.

1 Introduction

LLMs exhibit systematic failures when iterating on subjective work: sycophancy, overcriticism, overcompromise, authorship bias, scope drift, and context collapse. These failures persist across prompting strategies—even in code generation, iterative refinement leads to progressive degradation [2]. Self-Refine [4] demonstrates that single-agent critique-and-revise loops can improve outputs but provides no mechanism to detect when revisions make things worse. Huang et al. [5] show LLMs cannot reliably self-correct without external feedback, motivating our external judge panel.

We present autoreason, extending Karpathy's autoresearch [1] to subjective domains. Each iteration produces three competing versions—the incumbent (A), an adversarial revision (B), and a synthesis (AB)—evaluated by a blind judge panel. The method constructs a fitness function through the evaluation process itself, analogous to how science uses peer review where math can use proofs.

We tested two modes. **Single pass**: one cycle through critic → B → AB → judge (5 LLM calls). **Convergence loop**: repeats the cycle with a full judge panel until the incumbent survives consecutive challenges. As we show in Section 4.1, the single pass does not outperform simpler baselines—autoreason's value emerges from iteration.

2 Method

Critic. A fresh agent identifies problems in the current incumbent A. Finds problems only—no fixes. Optionally receives ground-truth data for fact-checking (Section 5).

Author B. A fresh agent revises A based on the critique. Sees the task, A, and the critique. No drafting history.

Synthesizer. A fresh agent receives A and B with randomized labels. Produces AB by taking the strongest elements of each.

Judge Panel. 3 fresh agents rank A, AB, B via Borda count (3/2/1 points). Randomized labels and presentation order. Conservative tiebreak: incumbent wins ties. Convergence at k=2 consecutive A wins.

Every role is a fresh, isolated agent with no shared context. The original task prompt anchors all evaluation. Exact prompts are provided in Appendix A.

2.1 Formal Framing

Let d_t denote the incumbent document at pass t. Each pass produces candidates $\{d_t, B(d_t), S(d_t, B(d_t))\}$ where B is the adversarial revision operator and S is the synthesis operator. A judge panel J defines a stochastic preference relation: $J(x, y) \in [0, 1]$ is the probability that x is ranked above y. The winner is selected by Borda aggregation over n judges:

$$d_{t+1} = \arg\max_{c \in \{A,B,AB\}} \sum_{i=1}^{n} (3 - r_i(c))$$

where $r_i(c)$ is judge i's rank for candidate c, with ties broken in favor of d_t.

This defines a Markov chain over document states. Convergence ($d_{t+1} = d_t$ for k consecutive passes) corresponds to an absorbing state. By the Condorcet jury theorem, if each judge independently identifies the better document with $p > 0.5$, panel accuracy increases with n. However, same-model judges share systematic biases, bounding effective independence below n.

The connection to PSRO [10] is direct: each pass adds a candidate to an empirical game. Our retroactive analysis (Section 4.8) validates this framing: preferences are fully

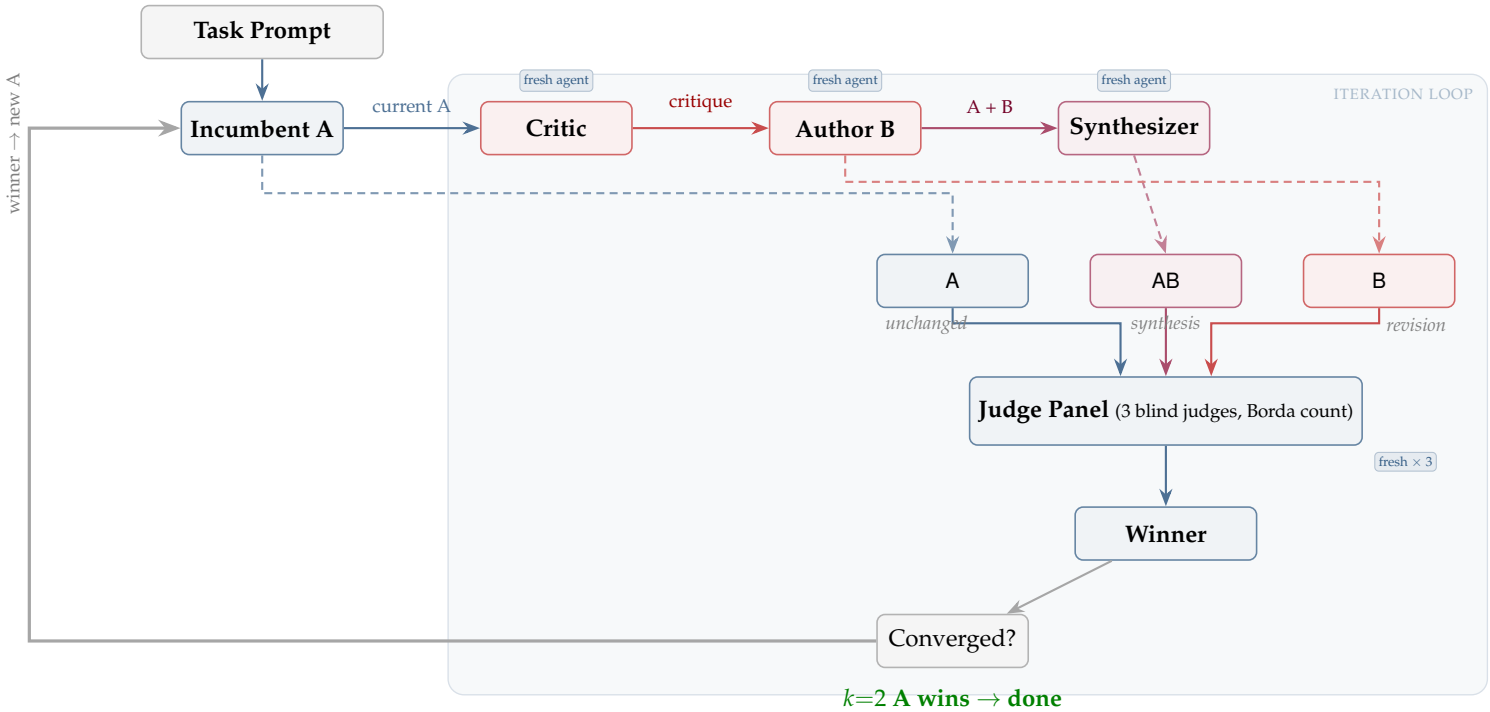

Figure 1. The autoreason loop. Each pass produces three candidates from the incumbent: the unchanged original (A), a synthesis (AB), and an adversarial revision (B). A blind judge panel selects the winner via Borda count. If A survives k=2 consecutive passes, the loop terminates. Every role is a fresh, isolated agent with no shared context.

transitive across 91 passes (1.1% Condorcet cycle rate), and Elo ratings plateau early, confirming the Markov chain reaches its absorbing state.

3 Experiments

All experiments used claude-sonnet-4-20250514 (temp=0.8 authors, 0.3 judges, max_tokens=4096) unless otherwise noted. Five tasks: (1) go-to-market strategy, (2) notification system design, (3) remote work policy, (4) competitive positioning, (5) incident response. Each task ran autoreason to convergence (k=2) plus four baselines (conservative, improve_this, harsh_critic, critique_and_revise) at 15 passes each from the same initial output, evaluated by a 7-judge blind panel. The paper-writing experiment (Section 5) used claude-opus-4.

4 Results

4.1 Single-Pass Evaluation

Method	Calls	T1	T2	T3	T4	T5
Harsh critic	1	**30**	**31**	23	**30**	**31**
Critique & rev.	1	**30**	30	**31**	28	23
Autoreason	5	24	22	24	17	24
Improve this	1	14	14	18	23	20
Conservative	1	7	8	9	7	7

Table 1. Single-pass Borda scores (max 35). Bold indicates winner per task.

Single-pass autoreason averaged 22.2 Borda, placing 3rd on 4/5 tasks. Harsh critic (29.0) and critique-and-revise (28.4) both outperformed it at 1/5th the cost. On a single pass, splitting work across five agents is not justified—simpler methods that concentrate effort into one strong revision win. **The question is whether iteration changes this.**

4.2 Multi-Task Comparison

Method	T1	T2	T3	T4	T5
Autoreason	**35**	19	**29**	25	**31**
Critique & rev.	13	**22**	26	**32**	19
Harsh critic	18	19	27	20	26
Conservative	21	19	15	18	15
Improve this	18	11	8	10	14

Table 2. Iterative Borda scores (max 35) across 5 tasks at 15 passes. Bold indicates winner. Autoreason won tasks 1, 3, 5; critique-and-revise won tasks 2, 4.

With iteration, autoreason won 3 of 5 tasks and placed second on the other two (avg Borda 27.8, avg rank 1.4). The iterative loop (avg 27.8) outperforms all single-pass methods (best: 29.0). The judge's protection against bad revisions compounds over passes—each iteration either improves the incumbent or leaves it unchanged, while baselines accumulate drift unchecked.

The pattern: autoreason excels on tasks with genuine trade-offs (strategy, policy, cross-timezone incident response) where the A/B/AB structure surfaces different valid perspectives. Critique-and-revise wins on tasks with concrete technical requirements (system design, competitive positioning) where a direct find-and-fix loop is more efficient.

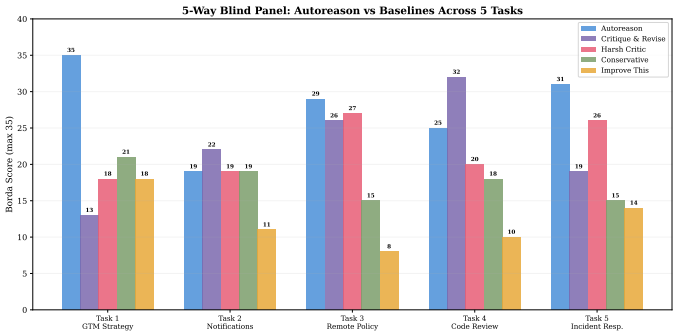

Figure 2. Per-task Borda scores for all 5 methods at 15 iterative passes. Autoreason (blue) wins tasks 1, 3, 5; critique-and-revise (purple) wins tasks 2, 4.

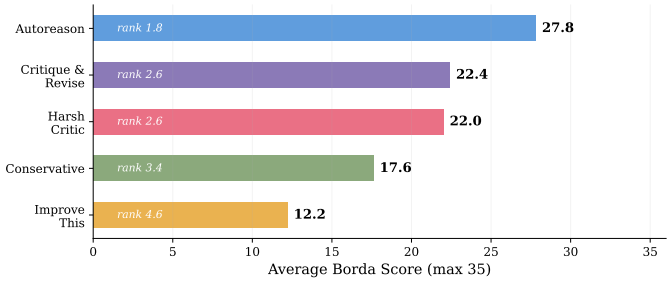

Figure 3. Aggregate performance across 5 tasks (iterative, 15 passes). Autoreason never placed below 2nd on any task.

4.3 Qualitative Improvement

The initial Task 1 output was a generic startup playbook: \$49/user pricing, \$100K MRR target with a 3-person team, no customer validation. After 14 passes, the converged output contained quantified pain (\$15K/incident $\times$ 6/yr), team-based pricing (\$1,499/mo matching the actual buying motion), validation from 50+ customer interviews with 75% pilot success rate, competitive positioning against named tools (OPA/Gatekeeper, ArgoCD, Flux), and unit economics (CAC \$2K, LTV \$54K, 90% gross margin). The adversarial process killed the unrealistic revenue assumptions.

4.4 Convergence Behavior

All 5 tasks converged at threshold k=2, requiring 9–28 passes. Task complexity correlates with convergence time: policy tasks converge faster (10 passes) than multi-stakeholder operational processes (28 passes).

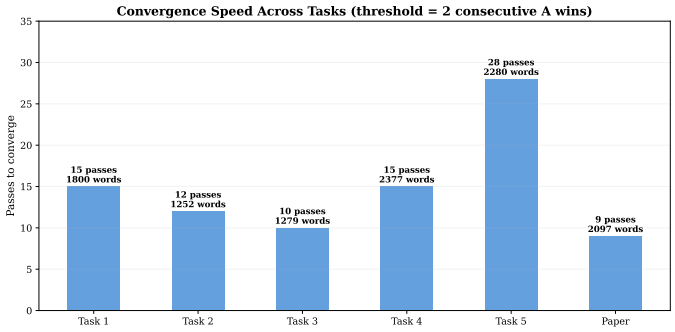

Figure 4. Passes to convergence across all tasks and the paper-writing experiment. Task 5 (incident response, 28 passes) required the most iteration; the paper (9 passes) the least.

4.5 Trajectory Analysis

Figure 5 shows the Task 1 trajectory as a tree, where vertical position represents cumulative drift from the initial version. A wins produce horizontal movement (no drift), B wins drift up by 1, AB by 0.5. At pass 15, A had won twice consecutively (convergence). But 11 more passes followed with B winning 5 of them. A 7-judge panel comparing pass 15 vs pass 25 chose pass 15 by 6–1—the later passes were churn, not improvement.

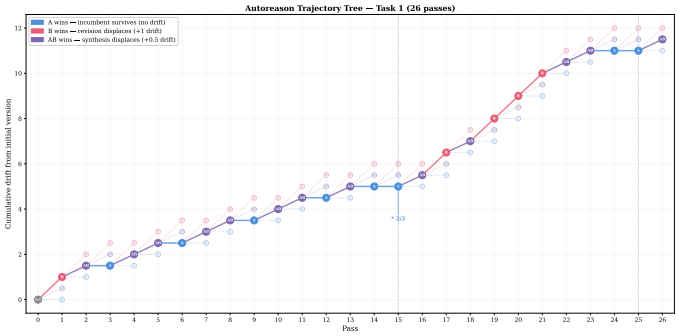

Figure 5. Trajectory tree, Task 1 (26 passes). Stars mark convergence points. Post-convergence activity is noise.

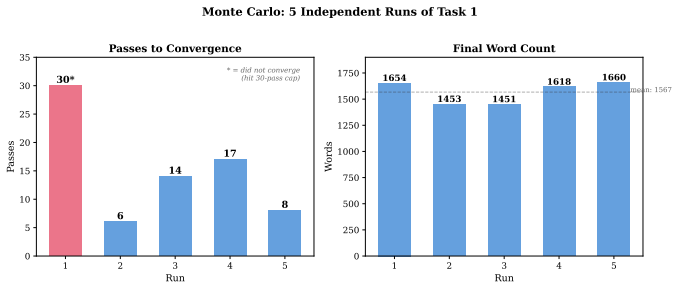

Figure 6. Monte Carlo analysis: 5 independent runs of Task 1. Left: passes to convergence (one run hit the 30-pass cap). Right: final word counts cluster tightly despite variable paths.

4.6 Chain-of-Thought Judges

Requiring judges to reason step-by-step before ranking dramatically improves convergence speed without changing the generation pipeline.

Judge Variant	Task 1	Task 2
Baseline (holistic)	14–15 passes	12 passes
Chain-of-thought	**5 passes**	**8 passes**
Decomposed (3 specialists)	7 passes	—

Table 3. Passes to convergence by judge variant. CoT reduces convergence 3× on Task 1.

CoT judges produce more decisive scores: A's winning scores were 8–9 (vs baseline's typical 6–7). The reasoning step acts as a debiasing mechanism—judges must articulate specific strengths before committing to a ranking, preventing verbosity bias and pattern-matching.

On Task 2 (autoreason's weakest task), CoT improved autoreason from Borda 19 (tied 3rd) to 25 (clear 3rd with 3 first-place votes), though critique-and-revise still won at 30. The task-dependent advantage persists but the gap narrows.

4.7 Monte Carlo Analysis

Five independent runs of Task 1 tested convergence reliability. 4/5 converged (80%). Convergence pass varied: 6, 8, 14, 17. One run hit the 30-pass cap without converging. Final word counts clustered tightly (1451–1660) despite variable path length, suggesting the loop reaches a similar quality ceiling regardless of path.

4.8 Retroactive Game-Theoretic Analysis

We analyzed all 91 passes across 5 tasks plus Monte Carlo runs:

Transitivity. Only 1 Condorcet cycle in 91 passes (1.1%). Judge preferences are almost perfectly transitive, confirming Borda aggregation is appropriate and the spoiler problem [11] is theoretical, not practical.

Elo ratings. Tracking incumbent Elo across Task 1 passes, ratings plateau at ∼1540 by pass 5–10. Elo at pass 15 (1547) vs pass 25 (1560)—barely different, confirming early convergence.

Pairwise dominance. Across all 5-way comparisons, autoreason wins 18/20 pairwise matchups (90%). The ordering is fully transitive: autoreason > critique-and-revise > harsh_critic > conservative > improve_this. No rock-paper-scissors dynamics.

4.9 Constrained Tasks

Task 1 with a constrained prompt (1000-word limit, 6 required sections with specific deliverables) ran for 25 passes (the maximum) without converging. B won 18 of 25 passes (72%). The constraint eliminated word count oscillation (all outputs 600–800 words vs 847–2500 unconstrained) but made convergence harder—specific constraints give the critic a verifiable checklist that always finds violations. Whether the loop would have converged given more passes is unknown.

4.10 Model Scaling: Sonnet 4.6

We ran autoreason on Task 2 using claude-sonnet-4-6 (a stronger model released after our main experiments) with CoT judges to test whether model capability improves convergence. The result was a complete reversal: autoreason came last.

Method	Borda	1st Place
Critique & rev.	**31**	3
Improve this	30	4
Harsh critic	23	0
Conservative	14	0
Autoreason	7	0

Table 4. Sonnet 4.6 on Task 2. Autoreason ran 50 passes without converging; all baselines ran 15 passes. CoT judges.

Over 50 passes, A won only 6 times (12%). AB dominated with 30+ wins; B took the remainder. The loop never approached convergence—A never won two consecutive passes. The stronger model produced AB syntheses so consistently preferred by judges that the incumbent could never survive a challenge.

This failure mode is structurally different from the bloat/prune oscillation observed with Sonnet 4. With the weaker model, A occasionally won on close calls and tiebreaks, creating windows for convergence. With Sonnet 4.6, both B and AB were reliably superior to A on every pass, and judges were decisive enough (scores of 8–9 for AB) to never let A through.

4.10.1 Remedy Experiments

We tested four modifications to recover convergence, all on the same setup (Sonnet 4.6, Task 2, CoT judges, 50-pass cap):

1. **Margin requirement**: AB/B must beat A by $\geq$2 Borda points to displace the incumbent; otherwise A survives.
2. **Scope-aware judges**: Judge prompt includes the initial version's word count and instructs judges to penalize unjustified expansion.
3. **Plateau detection**: Convergence triggers when A's average Borda score over the last 5 passes exceeds 6 (out of 9), regardless of consecutive wins.
4. **All three combined**.

Remedy	Passes	A	AB	B	Converged?
Margin	15	3	8	4	**Yes** (pass 15)
Combined	17	3	12	2	**Yes** (pass 17)
Scope-aware	35	4	25	6	No
Plateau	35	2	22	11	No
Baseline (no fix)	*50*	*6*	*30+*	*rest*	*No*

Table 5. Scaling remedy experiments (Sonnet 4.6, Task 2). Only experiments including the margin requirement converged.

The margin requirement is the effective fix (Figure **??**). Both experiments that included it converged (passes 15 and 17); both without it did not. In the combined experiment, the margin rule overrode 2 of 17 passes where AB would have won by 1 Borda point, creating the consecutive-win windows that the baseline could never produce. The margin-only experiment converged without any overrides—potentially stochastic, but the combined result confirms the mechanism.

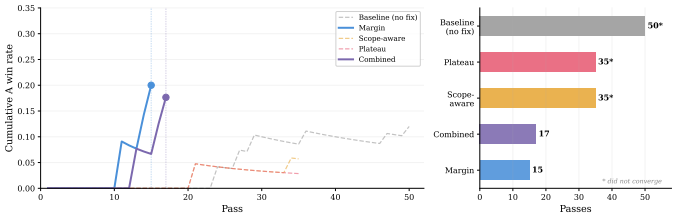

Figure 7. Scaling remedy results. Left: cumulative A win rate over passes (solid = converged, dashed = did not). Right: passes to convergence (* = did not converge, killed at cap). Only experiments including the margin requirement terminated.

Scope-aware judges had no effect: AB still won 25 of 35 passes (71%) despite explicit anti-bloat instructions. The stronger model's AB syntheses are preferred for reasons beyond verbosity. Plateau detection could never trigger because A's scores remained at 3–4 throughout—the incumbent is genuinely weaker per pass, not just losing on close calls.

The diagnosis is specific: the scaling failure is not about evaluation bias or stopping criteria. It is about the structural asymmetry between a static incumbent and fresh challengers with access to a critique. The margin requirement works because it raises the bar for displacement, giving the incumbent stability it cannot earn on pure quality alone. This is a design tradeoff—the margin rule accepts occasional false convergence (keeping a beatable incumbent) in exchange for eventual termination. Whether the converged output under the margin rule is actually good (as opposed to merely stable) requires a separate quality evaluation, which we leave to future work.

5 Writing This Paper

We ran autoreason on this paper using claude-opus-4 (all roles) with 3 Opus judges. All agents received actual experimental data as ground-truth context. Converged in 9 passes: AB→A→AB→AB→AB→AB→AB→A→A.

Ground-truth critic. Without ground-truth access, the initial Opus generation hallucinated a fabricated ablation study, fake confidence intervals, wrong model names, and incorrect role descriptions (Figure 7). With ground-truth access, the critic caught all four on pass 1. The author writes from understanding; the critic verifies against evidence—mirroring peer review.

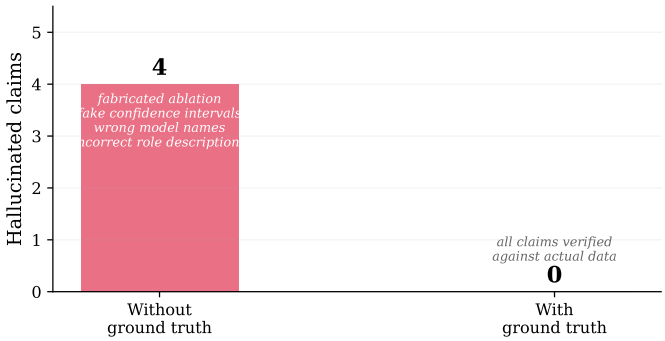

Figure 8. Ground-truth critic eliminates hallucination. Without reference data, the initial draft contained 4 fabricated claims. With ground-truth access, all claims were verified on pass 1.

Judge panel integrity. An earlier run with a mixed panel (Opus + Sonnet + Gemini) failed to converge for 11 passes because Gemini's output format failed our ranking parser, reducing the effective panel to 2 judges. Once fixed to 3 working judges, the same incumbent converged in 2 passes (Figure 8). Panel integrity is a prerequisite for convergence—a broken judge doesn't add noise, it prevents equilibrium.

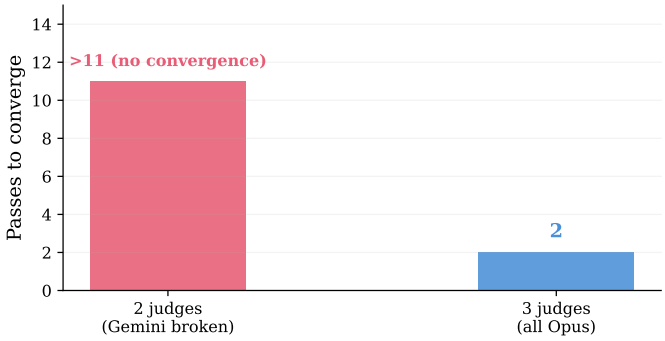

Figure 9. Judge panel integrity. A broken parser reduced the effective panel to 2 judges, preventing convergence for 11+ passes. Fixing to 3 working judges produced convergence in 2 passes.

6 Related Work

Autoresearch [1] uses val_bpb as an objective fitness function. Autoreason extends this to subjective domains. **SlopCodeBench** [2] shows iterative code degradation that prompting can't fix. **ACE** [3] documents context collapse in iterative rewriting. **Self-Refine** [4] demonstrates single-agent iterative improvement but Huang et al. [5] show self-correction is unreliable without external feedback. **AI Safety via Debate** [6] provides theoretical grounding for adversarial evaluation: truth has a natural advantage when agents argue opposing positions before a judge. **Constitutional AI** [7] uses principle-based self-evaluation as training signal. **LLM-as-Judge** [8] documents systematic biases (position, verbosity, self-enhancement) in LLM evaluation, motivating our randomization and panel design. **LLM Council** [9] uses multi-model panels for single-turn evaluation; autoreason adds iteration with role separation. **PSRO** [10] provides the game-theoretic framework: each pass adds a candidate to an empirical game.

7 Discussion

Task-dependent advantage. Autoreason is not universally superior. It excels where genuine tradeoffs exist between competing valid perspectives. For tasks with concrete, verifiable requirements, simpler methods suffice. The recommendation: use autoreason when you don't know which framing will produce the best result; use critique-and-revise when the improvement direction is clear.

CoT judges should be the default. The single cheapest improvement: adding "think step by step about each version" to the judge prompt. No architecture changes, 3× faster convergence, more decisive scores.

Convergence as practical stopping rule. The threshold does not guarantee optimality. It filters noise. Post-convergence passes on Task 1 produced no improvement (pass 15 beat pass 25, 6–1). The Monte Carlo results show 20% non-convergence at 30 passes, meaning some runs need a cap. The Elo analysis confirms quality plateaus early regardless of whether the consecutive-wins threshold is met.

Bloat/prune oscillation signals underdetermined scope. Constraining scope eliminates this but makes convergence harder by giving the critic concrete criteria to enforce. Figure 9 shows the structural story: autoreason grows steadily then stabilizes, while baselines either stagnate (conservative) or bloat monotonically (critique-and-revise).

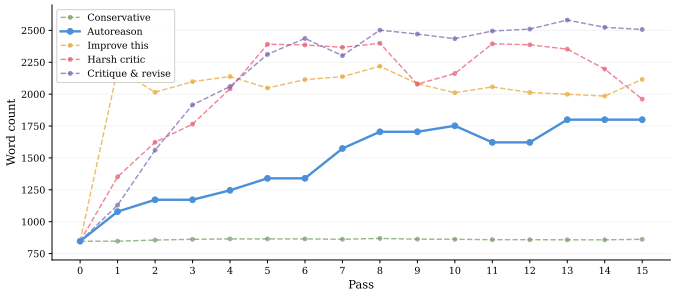

Figure 10. Word count trajectories across 15 passes. All methods start from the same 847-word initial output. Autoreason (solid blue) grows controllably; baselines either stagnate or bloat.

Model scaling is not free. The Sonnet 4.6 remedy experiments (Section 4.10) show that the scaling failure is addressable but only through the right mechanism. Scope-aware judging and plateau detection failed entirely—the problem is not evaluation bias or stopping criteria. The mar-

gin requirement works because it addresses the structural asymmetry: a static incumbent competing against fresh challengers with critique access will always lose close calls. Requiring challengers to win decisively compensates for this asymmetry. The tradeoff is accepting occasional false convergence in exchange for termination.

Limitations. All results from one model family (Anthropic). No human evaluation. No ablation studies isolating individual components. The constrained task experiment was capped at 25 passes—we cannot claim it would never converge. The paper-writing experiment used a different model (Opus) than the main experiments (Sonnet), limiting direct comparability.

8 Conclusion

Autoreason demonstrates that systematic isolation and blind evaluation can construct useful fitness functions for subjective domains. Across 5 tasks with Sonnet 4, it ranked first or second in every comparison. Chain-of-thought judging accelerates convergence $3\times$. Retroactive analysis confirms fully transitive preferences and early quality plateau. The method's advantage is clearest on tasks with genuine ambiguity, and its value requires iteration—a single pass does not justify the overhead. However, the Sonnet 4.6 scaling result shows this is not solved by default: stronger models break the convergence mechanism by producing syntheses that always outcompete the incumbent. A margin requirement—requiring challengers to win by ≥ 2 Borda points—recovers convergence, while scope-aware judging and plateau detection do not. The fix targets the right problem: the structural asymmetry between a static incumbent and fresh challengers, not evaluation bias. The primary insight remains: the obstacle to LLM self-improvement in subjective domains is not capability but evaluation methodology.

References

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[9] A. Karpathy. LLM Council, 2025.

[10] M. Lanctot et al. A unified game-theoretic approach to multiagent reinforcement learning. *NeurIPS*, 2017.

[11] K. Arrow. *Social Choice and Individual Values*. Wiley, 1951.

A Prompts

A.1 Critic

```
System: You are a critical reviewer. Your only job is to find real problems. Be specific
and concrete. Do not suggest fixes.
User: Find real problems with this proposal. Focus on: things that won't work as
described, complexity that doesn't pay for itself, assumptions that are wrong, missing
pieces. Do NOT propose fixes. Just the problems.
```

A.2 Author B

```
System: You are a senior consultant revising a proposal based on specific criticisms.
Address each valid criticism directly. Do not make changes not motivated by an identified
problem.
User: [TASK] + [VERSION A] + [CRITIC OUTPUT] Revise to address these problems. For each
change, state which problem it fixes.
```

A.3 Synthesizer

```
System: You are given two versions as equal inputs. Take the strongest elements from each
and produce a coherent synthesis. This is not a compromise.
User: [TASK] + [VERSION X] + [VERSION Y] (labels randomized)
```

A.4 Judge (Baseline)

```
System: You are an independent evaluator. You have no authorship stake in any version.
User: [TASK] + Three proposals (randomized labels and order). Rank best to worst.
RANKING: [best], [second], [worst]
```

A.5 Judge (Chain-of-Thought)

```
System: You are an independent evaluator. Think carefully before deciding.
User: [TASK] + Three proposals. For each, think step by step: 1. What does it get
right? 2. What does it get wrong or miss? 3. Are numbers and claims defensible? 4. Is
detail appropriate or bloated? After reasoning, rank all three. RANKING: [best], [second],
[worst]
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B Trajectory Trees

Figure 10 shows trajectory trees for all experiments.

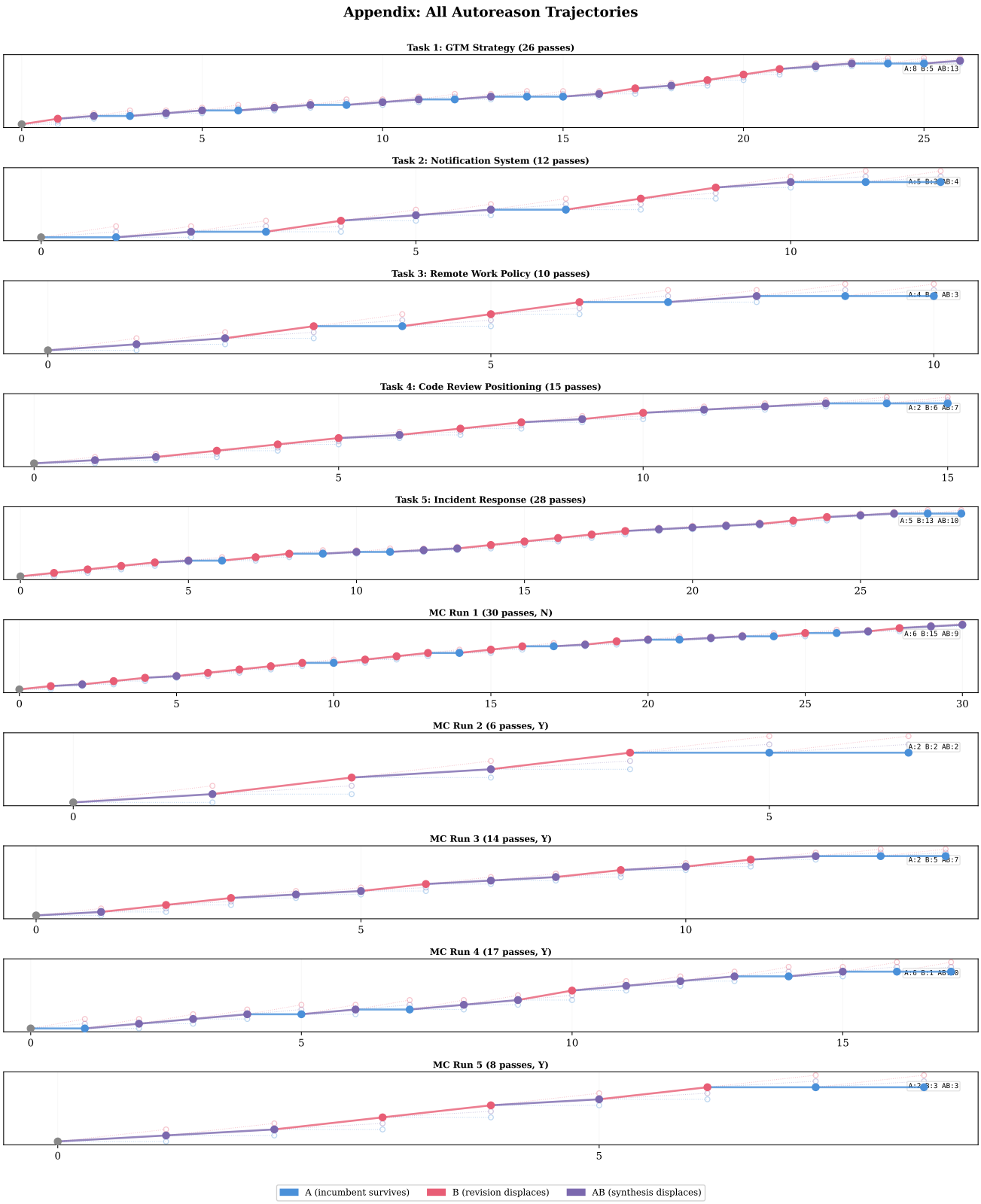

Figure 11. All trajectories. Tasks 2–3 converge quickly (10–12 passes). Task 5 requires 28 passes. Monte Carlo runs show variable convergence speed (6–30 passes) on the same task.